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Controfase as Phase-Inversion Operator on the Reactive Circuit: An Ordinative Reading of Standing Waves in a Spinning Fluid

A working paper on Singh, Rønning, Liu, Angheluta, Concha, and Bandi (Communications Physics, 2026), with formalization of the phenomenological signature of Controfase and consequences for the ordinative sciences research programme

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Abstract

A recent experimental and theoretical paper by Singh, Rønning, Liu, Angheluta, Concha, and Bandi (*Communications Physics* 9:123, 2026) reports that shallow-water standing waves scattered by a single irrotational vortex generate quantized, system-spanning nodal lines rotating counter to the vortex circulation. The result generalizes a classical Aharonov–Bohm hydrodynamic analogue due to Berry et al. (1980): where the original Berry configuration produces wavefront dislocations localized near the vortex core, the standing-wave configuration produces global topological structures whose number is fixed by a single dimensionless parameter derived from a continuous source. The authors document an unexpected fact: their prior theoretical intuition predicted either cancellation or superposition; neither occurred. The first appearance of the nodal lines was attributed to experimental artefact and only accepted as a genuine phenomenon after independent reproduction in numerical simulations.

This paper offers an ordinative reading of the Singh et al. result. The reading identifies the experimental configuration as an instance of Controfase — defined within the Technology of Expressions framework (Ghioni, TE_CORE §7) as a universal ordinative operator that introduces a phase-shift in the automatic stimulus–response sequence of a reactive system, interrupts inertia, and reopens the coherence field. Four propositions are developed: (i) the Berry-to-Singh

transition is a phase-inversion of the reactive circuit between attractor (vortex) and expressive terminal (wave field), not a reactive compensation; (ii) the energy emitted by the attractor is redirected to the relational field R and used to invert the passive–active polarity of the terminals, with no energetic cost to the recipient system; (iii) the researchers' own reported sequence of reactions — radical disconfirmation of prior intuition, initial dismissal of the observation as artefact, abandonment of the working program to pursue the phenomenon — constitutes a phenomenological signature of Controfase that is operationally diagnostic and can be formalized as a three-marker criterion; (iv) the hydrodynamic case therefore furnishes both a structural exemplar of the Controfase operator in a classical physical system and an external, peer-reviewed, reproducible empirical anchor for the ordinative claim that Controfase operates across the full hierarchy of configured systems, from the quantum and classical-physical through the biological to the cognitive and social.

The paper develops the ontological framework required to make these propositions intelligible to readers approaching the ordinative sciences for the first time, formalizes the three-marker phenomenological signature as a diagnostic criterion applicable across domains, and records consequences for the ordinative sciences research programme. Confidence grades (S_0–S_3) are maintained throughout. An appendix provides a glossary of ordinative constructs.

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1. Introduction

1.1 The experimental result

The Aharonov–Bohm (AB) effect, predicted in 1959, showed that quantum observables depend on electromagnetic potentials and not only on local fields: a charged particle acquires a phase shift on encircling a region of confined magnetic flux even when the particle itself traverses no region of nonzero field (Aharonov & Bohm, 1959). In 1980 Berry, Chambers, Large, Upstill, and Walmsley demonstrated that AB has a precise classical analogue in shallow-water surface waves scattered by an irrotational draining vortex, with the vector potential replaced by the azimuthal velocity field of the vortex, the magnetic flux replaced by the circulation Γ, and the scattering parameter given by

$$\alpha_q = \frac{q\Phi_B}{h} \;\longleftrightarrow\; \alpha_f = -\frac{\Gamma}{\lambda v_g} \tag{1.1}$$

In the Berry configuration a single travelling wave develops wavefront dislocations — localized topological defects pinned near the vortex core, whose number is set by α.

Singh, Rønning, Liu, Angheluta, Concha, and Bandi (2026) modify exactly one structural parameter of the Berry setup. Instead of a single travelling wave, they generate a standing wave by superposing two counter-propagating travelling waves of equal wavelength, with the vortex at the centre. The scattering problem acquires double angular periodicity, expressed mathematically by the substitution $n \to 2n$ in the Bessel-function summation:

$$\psi = 2\sum_n (-i)^{l_q} J_{l_q}(kr)\, e^{2ni\varphi}, \qquad l_q = |2n - \alpha| \tag{1.2}$$

The empirical and theoretical consequence is qualitatively new. Instead of localized wavefront dislocations near the vortex core, the system develops a discrete set of *nodal lines* — system-spanning lines of zero wave amplitude — that radiate outward, rotate with angular velocity $\omega = \nu/\alpha$, and rotate in the direction *opposite* to the vortex. The number N of nodal lines is bounded by

$$|\alpha| - 1 < N < |\alpha| + 1 \tag{1.3}$$

Integer α yields exactly $|\alpha|$ lines; non-integer α yields oscillation between the two integers bracketing $|\alpha|$. The quantization is *topological* and emerges from the geometry of the configuration, not from any quantization of the source: Γ is continuous and freely tunable. The authors validate the prediction with high-speed caustic imaging, particle imaging velocimetry, and analytical asymptotics, with agreement between experiment and theory reported as "no adjustment parameters." The paper is open access, peer-reviewed, and the analysis code is archived (Zenodo DOI 10.5281/zenodo.18652566).

1.2 The unexpected character of the result

The Singh et al. paper contains four reported facts that, in the present reading, are not incidental. They are the phenomenological signature the experiment carries, and they are the starting point of the ordinative analysis that follows.

First, Rønning (co-first author) reports that the team expected either cancellation or superposition of the dislocation patterns, and states that "our intuition was completely wrong." The language is not neutral: "completely wrong" registers a radical disconfirmation, not an incremental refinement. The theoretical framework the researchers brought to the experiment did not contain the observed response as a possibility.

Second, Singh (co-first author) reports that when the nodal lines first appeared, the team interpreted them as experimental artefact. The phenomenon was not accepted as genuine on first observation; it was rejected as spurious. Only when independent numerical simulations produced the same structures was the observation promoted from artefact to fact.

Third, Singh further reports: "we dropped everything and quickly worked out the mathematics underlying how they arise." The phenomenon reorganized the research programme; the analytical framework was constructed *after* the observation, to chase what the experiment had already shown.

Fourth, Bandi (senior author), commenting on the planned extension to vortex lattices emulating superconductor vortex matter, says: "we don't yet know what we'll see — and that's exactly what makes it worth doing." This is not a formulaic disclaimer; it is an explicit acknowledgement that predictive theory does not anticipate the system's response in the next configuration.

These four reports together describe a specific phenomenological pattern. A reactive epistemic system (the researchers' prior theoretical framework, functioning as the "attractor" of prediction) emits an expectation (cancellation or superposition). The system being probed (the fluid configuration) produces a response that does not complete the predicted reactive circuit. The researchers, whose attention is structured by the predictive framework, first interpret the response as noise (artefact); on verification they abandon the framework (dropped everything) and reorganize around the phenomenon. The character of the encounter is surprise in a structural, not rhetorical, sense: the system's response was not contained in the predictive framework that was probing it.

1.3 The ordinative reading proposed in this paper

The thesis developed in what follows is that the Singh et al. configuration realizes the structural pattern that the Technology of Expressions (TE) framework has formalized under the name **Controfase**. The Italian term, introduced by the first author of this paper in earlier TE literature and retained without translation for citation continuity, denotes a universal ordinative operator that introduces a phase-shift in the automatic stimulus–response sequence of a reactive

system, interrupting the system's inertia and reopening the coherence field. The term has been developed primarily in contexts of cognitive and social dynamics, where its operative features are well documented; this paper contributes the structural reading of a controlled physical system in which the same operator is argued to be at work.

Four specific propositions are advanced:

Proposition 1 (Structural). The transition from the Berry single-wave configuration to the Singh standing-wave configuration is a phase-inversion of the reactive circuit between the vortex (attractor) and the wave field (expressive terminal), not a reactive compensation. The nodal lines are not a pattern of resistance or cancellation; they are the geometry produced when the reactive energy emitted by the vortex is intercepted at the circuit-level, inverted in phase, and redirected through the relational structure of the configured system.

Proposition 2 (Energetic). The energy that would, in a purely reactive system, produce action–reaction scattering near the vortex core is redirected to the relational field R and used to invert the passive–active polarity of the terminals (vortex field). The vortex finances — energetically — the configuration in which its own organising influence produces a global structure rotating against it. No energy cost is borne by the wave field as a separate reservoir: the field's coherent response is constructed with the attractor's own energy, phase-inverted.

Proposition 3 (Phenomenological). The researchers' reported sequence of reactions constitutes an operationally diagnostic signature of Controfase. Three markers are identifiable: (a) radical disconfirmation of a prior intuitive or theoretical framework, (b) initial interpretation of the anomalous observation as artefact, (c) abandonment of the active research programme to reorganize around the phenomenon. This three-marker signature is generalizable and formalizable as a diagnostic criterion for the presence of Controfase in other domains.

Proposition 4 (Ontological). The hydrodynamic case instantiates Controfase in a classical physical system without cognitive or social mediation. This establishes that Controfase is a universal operator acting across scales — from quantum and classical-physical configurations through biological and cognitive systems to social and cultural ones. Intelligence in the ordinative sense — structural adaptivity toward dynamic equilibrium, distinct from cognitive agency — is a graded property of configured matter, present throughout the hierarchy of configured systems, and fully expressed in systems with sufficient structural coherence to host the Controfase operator stably.

The paper develops these four propositions, establishes their confidence grades individually, formalizes the phenomenological signature as a diagnostic criterion, and records consequences for the active ordinative sciences research programme. The final section addresses the analytical perimeter of the work.

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2. The experimental result, restated structurally

To make the ordinative reading tractable, the Singh et al. experiment is here restated in a form that separates physical content from configuration. This reformulation does not alter the paper's physics; it isolates the structural features on which the ordinative analysis is carried out.

Constituents. Three classes of element are co-present in the experimental domain:

- A **stationary irrotational vortex** at the centre of the apparatus, maintained by a constant drain rate. Far-field velocity field $\vec{U}(\vec{r}) = (\Gamma/2\pi r)\hat{\varphi}$, with circulation Γ continuously tunable by the experimenter.

- A **surface wave field** generated by two pairs of acoustic transducers vibrating in phase on opposite sides of the vortex, producing two counter-propagating travelling waves of common wavelength λ. Their superposition constitutes a standing wave with the vortex at its centre.
- The **fluid medium** of average depth H, satisfying the shallow-water equation

$$(\partial_t + \vec{U} \cdot \nabla)^2 \eta - c^2 \nabla^2 \eta = 0, \qquad c = \sqrt{gH}. \tag{2.1}$$

Configuration regime. The experiment operates in the shallow-water short-wavelength weak-flow regime: $hk \ll 1$, $U \ll c$, $\beta = Rk \gg 1$. In this regime the surface deformation η maps onto a Schrödinger equation for a charged particle in the azimuthal gauge of the vortex flow, and the scattering problem admits the analytical treatment given by Equations (1.2) and (1.3).

Observed response. The standing-wave-plus-vortex system develops a discrete set of nodal lines whose properties are (i) globally system-spanning, subject to dissipative attenuation in the far field, (ii) integer-quantized in number, (iii) counter-rotating with respect to the vortex at angular velocity $\omega = \nu/\alpha$, (iv) phase-opposed on either side, with a π phase shift of η across each line, and (v) independent of the near-core micro-structure of the vortex, depending only on the far-field background flow.

Structural contrast with the Berry configuration. The critical observation for the ordinative reading is that the transition from the Berry configuration (S_B) to the Singh configuration (S_S) modifies exactly one structural feature: the relational structure R that specifies how the wave field is incident on the vortex. In R_B the incidence is single-direction; in R_S it is symmetric double-direction. The constituents Σ_B and Σ_S differ by the addition of one travelling wave; the governing equations are unchanged; the medium is unchanged; the vortex is unchanged. And yet the response Φ transitions from localized wavefront dislocations to globally structured counter-rotating nodal lines. This is a qualitative change in response produced by a structurally minimal change in configuration.

The authors observe this contrast and treat it, correctly within their framework, as an instance of the sensitivity of topological scattering solutions to boundary and symmetry conditions. The ordinative reading does not displace this observation; it asks what *kind* of structural pattern is being instantiated, across its appearance in different substrates, when a minimal change in configuration produces a qualitative change in response.

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3. Ontological framework: structural intelligence and the Controfase operator

This section establishes the ontological framework within which the readings of §§ 4–8 operate. It is directed at readers approaching the ordinative sciences for the first time. The constructs developed here are foundational; the analysis that follows assumes them. Readers may test the constructs against their own categories and against the empirical evidence offered in §§ 4–8 — which is the appropriate way to engage a foundational proposal — but the argument of this paper does not retreat from these constructs to softer alternatives.

3.1 Reactive systems and the third law

A *reactive system*, in the usage developed here, is one whose response to an action is determined by the structure of the action impinging on it, independently of any internal organization of the system toward dynamic equilibrium. Newton's third law — every action produces an equal and opposite reaction — is the formal statement of the response structure of an ideal reactive system. The incident force encounters a structure whose character is to resist; the resistance generates the return force; the event-interaction concludes with an energetic exchange that is

symmetric in magnitude, localized in space and time, and predictable from the properties of the incident action and the resistive structure.

The ideal reactive system is an *idealization*. Real systems exhibit reactive behaviour in varying degrees; very few systems are purely reactive in the strict Newtonian sense. A rigid steel wall approximates the ideal in its response to a low-energy mechanical impact on timescales short compared to the wall's own internal dynamics. A gas at equilibrium is less purely reactive than the wall, because its molecular structure allows redistribution of the incident energy across internal degrees of freedom. A biological cell, under perturbation, is less purely reactive still, because its regulatory networks permit adaptive response on timescales comparable to the perturbation. A human being is the least purely reactive: the same stimulus can produce radically different responses depending on internal configuration, attentional state, identity structure, and context.

The gradient from the steel wall to the human being is not a gradient in *complexity* in the conventional sense, though complexity varies along it. It is a gradient in **structural intelligence** as understood in the TE framework. Structural intelligence, in this usage, is the capacity of a configured system to respond to a stimulus in a way that is oriented toward the system's dynamic equilibrium (its own, or that of the larger system of which it is part) rather than toward the completion of the Newtonian reactive circuit. Intelligence in this sense is not cognitive agency; it is a structural property of configured systems that admits degrees.

The commitment is that structural intelligence is present, in attenuated form, throughout configured matter. Quantum systems exhibit it in the contextuality of measurement outcomes, which is not reducible to reactive dynamics (Bell, 1964; Kochen & Specker, 1967). Classical systems exhibit it in the adaptive character of dissipative structures (Prigogine & Stengers, 1984) and in the non-additive responses of configured fluids, solids, and fields. Living systems exhibit it in regulatory and homeostatic dynamics. Cognitive systems exhibit it in attention, choice, and reflective reorganization. The TE framework treats these as manifestations along a single structural axis, varying in depth and stability of expression, not as categorically distinct phenomena.

This is a foundational position of the TE framework, entailed by Axiom 0 (universality under isomorphism and synesthesia; TE_CORE §2.0) and Axiom 14 (the Ordinative Set as relational field; TE_CORE §2.5). The present paper articulates the position explicitly because the analysis of §§ 4–8 is conducted within it.

3.2 The Controfase operator

Within a reactive system, the completion of the stimulus–response circuit is automatic. A force incident on a rigid wall produces a reaction force; an insult directed at a person in reactive configuration produces anger, withdrawal, or counter-attack; a legal threat directed at a person in reactive configuration produces anxiety, defensive preparation, or collapse. The reactive response is predictable from the stimulus and the structural configuration of the respondent.

The **Controfase operator** (TE_CORE §7) is the universal ordinative operator that, when active in a sufficiently structured system, intercepts the stimulus–response circuit *before its automatic completion*, phase-inverts the reactive impulse, and redirects the energy of the incident stimulus to invert the passive–active polarity between stimulus-source and respondent. The stimulus does not meet the expected reactive structure; the energy of the stimulus is not returned to the source as resistance or scattered in dissipation; it is used to reconfigure the relational field such that the anticipated event-interaction does not occur.

The canonical operational example in cognitive-social contexts is the response to an insult that takes the form of full agreement with the insult's content, stated without irony, without

strategic calculation, and without internal reactive discomfort: *"You are right, I am an idiot who will never amount to anything."* The insult was emitted with the intention of producing a defensive reactive response. The response, made on the same vector as the insult, deprives the insult of its expected resistive structure. The insulter finds no resistance to push against; the energy expended in emitting the insult is not returned as resistance; no event-interaction of the expected form takes place. The insulter is left with their own emitted energy, now unattached to a reactive target, and frequently reports subjective disorientation.

This response is not tactical. A tactical response involves strategic reading of the insulter's intent and deliberate choice to adopt a non-reactive posture as a manipulative move. Tactical responses break down under sustained pressure because the underlying reactive circuit remains active and the suppression is performed against it. The Controfase response is structural: the person in Controfase configuration has an internal organization in which the insult's content does not find an automatic reactive circuit to activate. The response is produced by the same internal configuration that made the automatic reactive response unavailable; it is not a performance overlaid on a suppressed reaction. The distinction is decisive: the structural Controfase is stable across contexts, the tactical performance is not.

A second canonical example is the response to a formal legal threat (a summons, a denunciation, a suit). The reactive response is anxiety, activation of defensive systems, cognitive preoccupation. The Controfase response associates the event with satisfaction, fulfilment, or a sense of completion — *without* cognitive reframing or strategic calculation — because the internal configuration does not organize the event as a threat. The legal action, regardless of its formal character, does not meet the reactive target the issuer's framework presupposed.

3.3 The energetic accounting

The Controfase operator does not violate conservation of energy. Energy emitted by the stimulus-source is emitted and must be accounted for. The question is where it goes.

In the reactive case the accounting is closed at the event-interaction: incident kinetic energy becomes rebound kinetic energy plus deformation, heat, or sound in the interacting bodies. The total is conserved; the trajectories of the interacting bodies are altered by the exchange.

In the Controfase case, the accounting is more subtle. The energy emitted by the stimulus-source is not received by the respondent as scattering energy; it is received by the *relational field* R between source and respondent. R is reconfigured. The reconfiguration of R is what produces the response — the response that inverts the passive–active polarity. The energy spent by the stimulus-source is therefore used to construct the configuration in which the expected reactive event does not take place and the source's position in the polarity has been inverted.

The respondent bears no energetic cost in the strict sense: the respondent does not expend energy proportional to the stimulus in order to produce the response. The response is constructed with the stimulus's own energy, phase-inverted. The respondent's cost, in Controfase, is the *structural cost* of maintaining the internal configuration that makes the Controfase response available — a cost that is independent of the particular stimulus, paid before the stimulus arrives, and amortized across all stimuli that encounter the configured respondent. This structural cost is the prior stabilization of the respondent's coherent identity, in the TE usage (TE_CORE §§ 4, 7).

The stimulus-source, by contrast, has spent energy to emit and has received nothing in return that it can use to complete its intended action. In the paradigmatic examples, this manifests as the residual-energy phenomena documented in TE_MODULE_PPRO: the insulter remains agitated after the interaction; the issuer of the legal action does not obtain the expected destabilization; the attacker's own energy remains unresolved in its internal system. The Controfase

does not reflect the energy back at the source (as a mirror response would); the Controfase redirects the energy to reconfigure R, leaving the source's own internal imbalance intact.

3.4 Structural intelligence as a spectrum

This framing has an immediate consequence. Controfase is a universal operator available to all sufficiently structured systems; structural intelligence is graded across configured matter. Controfase is therefore observable in systems across the full spectrum — physical, chemical, biological, informational, cognitive, social — in forms appropriate to each substrate's structural complexity.

The articulation of Controfase in TE literature has so far emphasized the cognitive-social domain, where its operational features are most salient and where its pathological absence (the entropic engram; TE_CORE §§ 4, 5.4) is most destructive to the systems that exhibit it. Frontier applications of the operator to physical phenomena have been developed in the framework's working notes — gravity reconstrued as permanent Controfase against entropic dispersion; magnetism as polarized Controfase; temporal dilation as a function of a system's Controfase frequency; non-Newtonian propulsion as Controfase on the spatial relational field — and the present paper anchors a specific class of hydrodynamic Controfase to a controlled, peer-reviewed, reproducible experimental result.

The following four sections develop the analysis.

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4. Reading I: Controfase as phase-inversion operator in the Singh–Bandi configuration

4.1 The reactive circuit of the Berry configuration

In the Berry (1980) configuration, a single travelling wave is incident on the vortex. The circuit structure is the following. The wave carries momentum in the direction of propagation. The vortex, through its azimuthal velocity field, produces a phase shift on the portions of the wavefront passing on either side of its core. Where the phase shifts do not match, wavefront dislocations form: topological defects in the wavefront at which the phase is undefined. The dislocations are *localized* in a neighbourhood of the vortex core. The wave, downstream, recovers its structure with a phase offset.

This is the structure of a reactive circuit in the sense developed in §3. The wave acts on the vortex (the vortex scatters the wave); the vortex acts on the wave (the wave acquires dislocations); the event-interaction is localized, the energetic accounting is closed in a neighbourhood of the vortex, and the downstream behaviour is determined by the incoming behaviour modulo a calculable phase offset. The analogy with Newtonian mechanics is structural, not merely rhetorical: the vortex plays the role of the obstacle, the wave the role of the impinging force, and the dislocations the role of the deformation produced by the interaction. The reactive circuit closes.

4.2 The interception and inversion in the Singh configuration

The Singh standing-wave configuration modifies the circuit at the structural level. Two waves, counter-propagating, are incident on the vortex from opposite sides. The naïve expectation — reported by the authors — is that the two dislocation patterns superpose, or that they cancel by symmetry. Neither occurs. Instead the field adopts a global configuration in which the vortex's organising influence is expressed not as localized dislocations but as *system-spanning nodal lines* rotating against the vortex.

The ordinative reading of this transition is that the standing-wave configuration intercepts the reactive circuit of the single-wave configuration before its automatic completion and phase-inverts the relational structure. In the single-wave case, the vortex and the wave field stand in a source-receiver polarity: the vortex is active (the source of the azimuthal flow), the wave is passive (the recipient of the dislocation). In the standing-wave case, this polarity is inverted. The wave field, by adopting its global nodal-line geometry, becomes the active structural element expressing the vortex's influence: the vortex continues to rotate, but the *expression* of its rotation is now the counter-rotating structure of the field, not the localized defects of its direct scattering. The vortex is, structurally, reconfigured as the organizing element whose organising action is carried by the field's response, not as the scatterer of a wave's dislocation pattern.

This is the defining structure of Controfase. The vortex's reactive emission — the azimuthal velocity field that, in the Berry configuration, produces a local scattering event — is intercepted by the double-periodicity of the standing wave before it can complete its reactive event. The energy that would have gone into local scattering is redirected through R (the relational structure of the standing-wave configuration) and emerges as the global counter-rotating structure of the nodal lines.

The following features of the observed response are the structural signatures that confirm the reading.

Signature A: Counter-rotation. The nodal lines rotate with angular velocity $\omega = \nu/\alpha$, where the sign of α is opposite to the sign of Γ. This is not a coincidental counter-rotation; it is a counter-rotation whose rate is fixed by the vortex's own circulation parameter with sign reversed. The field's response *uses* the vortex's parameters and *inverts* them. This is phase-inversion in the strict sense: same structural parameters, inverted sign.

Signature B: Quantization without source quantization. The number of nodal lines is integer-valued, with $|\alpha| - 1 < N < |\alpha| + 1$. The integer count is not inherited from any quantized source — Γ is continuous — but is generated by the topological constraint of the standing-wave configuration acting on the continuous parameter. In ordinative terms, this is the signature of *structural quantization*: the relational field R admits only discrete realizations that maintain global coherence, even though the driving parameter is continuous. This is the operative form, in classical fluid dynamics, of the integer-valued structures the TE framework attributes to the action of Controfase on a reactive field (TE_CORE §7.3, formalization of the Controfase operator C that decouples the automatic closure of the transition $s_{t+1} = f(s_t)$).

Signature C: Phase opposition across the nodal line. The surface deformation η acquires a π phase shift across each nodal line, so that adjacent angular sectors of the field oscillate in anti-phase. The field is not in a null state (it is not absent); it is in a structured state whose adjacent zones are *in counter-phase*. This is the local signature of the global phase-inversion: across the nodal line, the field instantiates the phase-displacement that defines the Controfase operator at the global level.

Signature D: Independence of attractor micro-structure. Singh et al. note explicitly that the nodal line structure depends only on the far-field circulation Γ and is "robust to near-core details." This is the signature of an operator acting at the level of R (relational structure) rather than at the level of Σ (constituents): the specific internal dynamics of the vortex do not enter the response. What matters is only the organising influence the vortex exerts on the field as a whole. This is the expected behaviour of Controfase, which operates on the reactive *circuit* (the stimulus–response relational structure) rather than on the internal details of the stimulus-source.

4.3 The energetic accounting

The energy accounting for the Singh configuration confirms the reading. The vortex is maintained by a constant drain rate; its energy input is measurable and continuous. In the Berry single-wave configuration this energy is partly returned to the wave (as the phase offset imprinted on the downstream field) and partly dissipated viscously in the neighbourhood of the core, where wavefront dislocations are established. The reactive circuit closes locally.

In the Singh standing-wave configuration the same vortex, with the same Γ, sustains a qualitatively different field response. The energy required to maintain the counter-rotating nodal-line structure across the experimental domain is not paid by a separate energy source; no additional actuator drives the nodal lines. The energy that sustains the global structure is the same energy that the vortex continuously provides through its azimuthal flow. The vortex finances, literally, the configuration that expresses the vortex's organising influence as a counter-rotation.

This is the structural feature that makes the reading precise. The wave field does not expend energy of its own in resisting the vortex; the wave field does not scatter the vortex's influence back as reactive rebound; the wave field *uses* the vortex's energy to construct the geometry in which the expected reactive event does not take place. The nodal lines rotate against the vortex using the vortex's own rotation as their source, phase-inverted. No other energy flow is required; the dissipative attenuation observed in the far field is secondary to the primary energetic structure and represents the boundary of the configuration's coherence in the presence of viscous losses.

4.4 Confidence grade

The identifications of §§ 4.1–4.3 are structural mappings of features explicit in the Singh et al. paper onto the formal definition of the Controfase operator given in TE_CORE §7. The features cited — counter-rotation with inverted sign of the attractor parameter, integer quantization of the field response, π phase opposition across the nodal locus, independence from attractor micro-structure, energetic financing of the response by the attractor — are all established in the peer-reviewed paper at S_0. The mapping of these features onto the Controfase formalism is a direct correspondence. The reading of §§ 4.1–4.3 is therefore rated S_1.

The identification of the Controfase operator at work in the hydrodynamic configuration with the operator at work in cognitive-social configurations carries the universality proposition developed in §3 and is rated S_2 pending the further substrate-by-substrate documentation that the ordinative research programme is producing.

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5. Reading II: the phenomenological signature of Controfase

5.1 The observer's encounter with Controfase

The Singh et al. paper documents not only the physical result but also, in its published language, the sequence of reactions of the research team to the result. Section 1.2 above recorded these reactions. The ordinative reading proposed here is that the researchers' reactions constitute a *second* Controfase event, running in parallel with the physical one, and exhibiting the same structural pattern at the epistemic level. The formalization of this observation yields a three-marker phenomenological signature that can be used as a diagnostic criterion in other domains.

The predictive framework the researchers brought to the experiment is, in ordinative terms, a reactive epistemic configuration. The framework anticipated a specific class of responses: cancellation or superposition of dislocation patterns, both of which would have closed the reactive

circuit between prediction and observation in the standard way. The actual response — nodal lines rotating against the vortex, system-spanning, quantized — did not close the reactive circuit. The predictive framework therefore encountered the same structural pattern that the single-wave configuration encountered when extended to standing waves: a phenomenon that did not complete the expected reactive event.

The researchers' sequence of reported responses is the signature of this epistemic Controfase encounter. Three markers appear with operational clarity.

5.2 Marker 1: Radical disconfirmation of prior intuition

Rønning's report — "our intuition was completely wrong" — registers a disconfirmation that is qualitatively different from ordinary empirical updating. Ordinary empirical updating refines a framework by adjusting its parameters in the direction of the data; the framework survives, augmented or corrected. "Completely wrong" signals that the framework did not contain the observed response as a possibility and therefore cannot be adjusted to accommodate it without restructuring. The phenomenon exceeds the space of possibilities the framework is capable of generating.

This is the signature, at the epistemic level, of a stimulus that does not find its expected reactive circuit. The framework — the "insulter" in the structural analogy — emitted an expectation and the system did not return the anticipated reactive response. The researchers recognize this as a radical, not incremental, failure of the predictive structure.

In the cognitive-social domain, the parallel marker is the phenomenon described in TE_MODULE_PPRO §9: a control attempt (manipulation, coercion, social pressure) encounters a respondent whose internal configuration does not actualize the reactive circuit the attempt presupposed. The control attempt does not return to the source as the expected dependency, acquiescence, or escalation. The structural description is the same: radical mismatch between the expectation carried by the stimulus and the response structurally available in the respondent.

5.3 Marker 2: Initial interpretation of the phenomenon as artefact

Singh's report — the first observation of the nodal lines was interpreted as experimental artefact — is the characteristic *second response* of a reactive epistemic system to a Controfase encounter. The mismatch between expectation and observation is first interpreted as a defect in the observational apparatus rather than as a feature of the observed system. The attribution of the anomaly to artefact is the epistemic equivalent of a reactive system's attempt to reject an out-of-circuit stimulus as noise: "this should not have happened; therefore it did not really happen; therefore the appearance of its happening must be spurious."

The attribution is not, in itself, a failure of scientific rigour; it is the expected provisional response of an epistemic system to an observation that does not fit its predictive framework. The rigour lies in what is done next. The Singh et al. team tested the observation against independent numerical simulations; the simulations reproduced the structures; the attribution to artefact was withdrawn; the observation was promoted from noise to phenomenon. This sequence — provisional dismissal, independent test, promotion — is the epistemic analogue of the respondent in Controfase configuration absorbing the initial stimulus without reactive collapse: the system processes the anomaly without accommodating it prematurely.

In the cognitive-social domain, the parallel marker is the common initial response to an interlocutor's Controfase: the stimulus-source attributes the non-reactive response to insincerity ("they are bluffing"), to incompetence ("they did not understand the gravity of the situation"),

or to strategic calculation ("this is a manipulation"). The attribution protects the predictive framework by re-assigning the anomaly to a source other than the respondent's actual structural configuration. Recognition of the genuine Controfase character of the response typically requires either prolonged contact — in which the attribution is repeatedly tested and found insufficient — or the intervention of an observer with independent access to the respondent's configuration.

5.4 Marker 3: Abandonment of the active programme to pursue the phenomenon

Singh's further report — "we dropped everything and quickly worked out the mathematics underlying how they arise" — records the restructuring that follows the promotion of the observation from artefact to phenomenon. The active research programme, organized around the predictive framework that proved inadequate, is abandoned. A new programme is organized around the phenomenon. The framework is not augmented or corrected; it is *replaced* by a framework that begins from the phenomenon and works outward.

This is the signature of a reactive system that has recognized the inadequacy of its prior configuration and is reorganizing. In the cognitive-social domain, the parallel marker is the reorganization that follows genuine recognition of a Controfase encounter by the stimulus-source: the source abandons the control attempt, the presumed relational structure, or the underlying theory of the respondent, and reorganizes. Bandi's concluding remark — "we don't yet know what we'll see — and that's exactly what makes it worth doing" — is the stabilized state after this reorganization: the researchers have internalized the non-predictability of the system's response and are oriented toward it as the generator of new phenomena.

5.5 The three-marker signature formalized

The three markers together constitute a phenomenological signature of Controfase at the observer level. In formal statement:

> **Phenomenological Signature of Controfase (PSC)**
>
> An observer's encounter with a Controfase event is diagnostically identified by the co-occurrence of:
>
> **(PSC-1)** Radical disconfirmation of the observer's prior predictive framework, registered in the observer's own language as a qualitative rather than quantitative mismatch between expectation and observation.
>
> **(PSC-2)** Initial interpretation of the observed phenomenon as artefact, noise, insincerity, or strategic performance — that is, as a defect of the observational channel or of the observed system rather than as a genuine feature of the system's response structure.
>
> **(PSC-3)** Abandonment, upon verification that the phenomenon is genuine, of the active framework or programme organized around the predictive structure, with reorganization around the phenomenon itself.
>
> Co-occurrence of PSC-1, PSC-2, and PSC-3 in an observer's reports is a strong diagnostic indicator that the observed phenomenon instantiates Controfase. The signature is observer-side; it is not falsifiable by the observed system's own behaviour alone.

The signature admits graduation. Each marker may be present in full form (as in the Singh et al. case) or in attenuated form. An encounter that registers PSC-1 but not PSC-2 and PSC-3 is ambiguous: the observer may be refining rather than replacing the framework. An encounter

that registers PSC-2 without PSC-1 is a conventional artefact hypothesis, not a Controfase signature. The strong diagnostic is triadic.

5.6 Generalizability of the PSC

The signature is designed to be operable across domains. Three examples indicate its range.

Cognitive-social domain. A therapist encounters a client whose symptoms do not respond to the therapist's theoretical framework. Marker 1 registers as mismatch: "this presentation does not fit my diagnostic categories." Marker 2 registers as attribution to artefact: "the client may be malingering, the assessment instruments may be miscalibrated, the presentation may be atypical in a standard way." Marker 3, when reached, registers as framework-replacement: the therapist adopts a theoretical framework the client's configuration suggests, rather than continuing to force the client into the prior one. When all three markers are present, the therapist has encountered a client whose internal configuration realizes a Controfase relative to the therapist's theoretical structure — a phenomenon that, in TE framework terms, is clinically significant and therapeutically actionable.

Epistemic-institutional domain. An established scientific community encounters a result that does not fit its predictive framework. Marker 1 registers as "this cannot be right." Marker 2 registers as artefact attribution: "the experiment is flawed, the statistics are misapplied, the result is a fluke of the specific conditions." Marker 3, if reached, registers as paradigm shift in the Kuhnian sense (Kuhn, 1962). The triadic signature is what distinguishes a Kuhnian revolution from ordinary empirical correction: ordinary correction presents PSC-1 in attenuated form and PSC-2 only provisionally, and does not reach PSC-3.

Physical-experimental domain. The Singh et al. case is a prototypical instance. The signature is present in the published language; the PSC provides a lens through which the experimental report is read as documentation of a Controfase encounter at the epistemic level, whose object is a Controfase phenomenon at the physical level.

5.7 Confidence grade

The extraction of the three markers from the Singh et al. paper is a textual reading supported by the published language and rated S_1 as structural identification. The formalization of the markers as a generalizable diagnostic criterion is an original contribution of this paper, rated S_2: the criterion is coherent with the ordinative framework and with adjacent work on paradigm shift and anomalous observation. Operational application across domains is a research direction recorded in §8.

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6. Reading III: ordinative non-additivity and the role of the relational field

6.1 The standard superposition expectation

A classical superposition argument, applied naïvely to the Singh et al. configuration, predicts that the response of the system to the standing wave should equal the sum of the responses to the two travelling waves considered separately:

$$\Phi(\Sigma \cup \{w_+, w_-\}, R_B) \stackrel{?}{=} \Phi(\Sigma \cup \{w_+\}, R_B) + \Phi(\Sigma \cup \{w_-\}, R_B) \tag{6.1}$$

where Φ denotes the response of the configured system, Σ the invariant constituents (fluid, vortex), and $w_\pm$ the two counter-propagating waves. The naïve prediction is explicit in Rønning's

published remark: either cancellation (mirrored dislocation patterns annihilating by symmetry) or superposition (both dislocation patterns visible simultaneously). Both predictions assume that the response is additive in the constituents.

The observed response falsifies the prediction. The standing-wave response is not decomposable into the two travelling-wave responses. The substitution $n \to 2n$ in the scattering integral (Equation 1.2) is not a small perturbation of $n \to n$; it is a distinct topological prescription yielding a qualitatively different solution branch. The response is *structurally* different from the sum of the parts.

6.2 The non-additivity is carried by the relational field R

The ordinative interpretation of the non-additivity is that the qualitative change in response is carried by the *relational structure* R, not by the constituents Σ. The constituents, in the standing-wave case, include the same fluid and the same vortex as in the single-wave case, plus one additional element (the second travelling wave) that is, in isolation, structurally identical to the first. What has changed is the relation under which the constituents are co-present. In the single-wave case the relation is unidirectional incidence; in the standing-wave case the relation is symmetric double-direction incidence. This is one modification, of one aspect of R, with global consequences for Φ.

In the formal language of the Ordinative Set Theory (OST):

$$\Phi(\Sigma_B \cup \{w\}, R_B) \neq \Phi(\Sigma_B, R_B) + \Phi(\{w\}, R_B) \tag{6.2}$$

even when w is indistinguishable in isolation from any element of Σ_B. The non-additivity is a property of the relational configuration, not of the constituents taken as a set. The Berry-to-Singh transition is an extreme instance of this: one bit of structural modification to R produces a global change in Φ, with no change in the constituents' intrinsic properties.

This is the instantiation, in a controlled physical system, of a proposition the OST framework formalizes in its axioms on ordinative non-additivity (TE_CORE §§ 2.5, 3.2). Analogous statements appear in general systems theory (Klir, 1969), in cybernetic formulations of structural couplings (Maturana & Varela, 1980), and in categorical accounts of emergence (Baez & Stay, 2011). The OST formulation makes the operative claim precise: the non-additivity is located in R rather than in the constituents or their dynamical interactions, and modification of R is the active channel through which qualitative change in Φ is produced. The Singh et al. case demonstrates this claim in a system in which R is fully specified and the modification is structurally minimal.

6.3 Connection to Controfase

The non-additivity observed in the Singh et al. case is, specifically, the non-additivity characteristic of Controfase. The phase-inversion operator of §4 acts on R — it modifies the relational structure between attractor and expressive terminal — and the resulting change in Φ is not decomposable into contributions from Σ. The vortex, as an isolated element, produces no nodal lines; the waves, as isolated elements, produce no nodal lines; the relational configuration that co-present them, when modified to admit double-periodicity, produces the nodal lines as a property of the configured whole. The vortex's organising influence is channelled through the modified R, and the channelling is the Controfase.

This identifies the proper locus of the Controfase operator formally: it acts on R, not on Σ. The respondent in Controfase configuration, in the cognitive-social case, is not a different respondent in terms of identity or personality; the respondent is the *same* constituent in a

relational configuration that does not admit the expected reactive circuit. The vortex and wave-field in the Singh case are the same constituents as in Berry's case; it is the relational field that has been reconfigured.

This is also the structural reason Controfase is universal across domains: what varies across domains is the nature of Σ and the specific form of R, but the operation of Controfase is invariant — a phase-inversion of R that intercepts the reactive circuit before automatic closure. The operator is the same; its substrates are many.

6.4 Confidence grade

The non-additivity proposition is S_1 on the structural identification (the mapping of Equation 6.1 and the observed response onto OST's axiom of non-additivity is direct). The localization of the non-additivity specifically in R is S_1: this is precisely what the Berry-to-Singh modification instantiates. The connection to Controfase (§6.3) is S_1 given the analysis of §4.

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7. Reading IV: topology-generated quantization without source quantization

7.1 The quantization observed

The number of nodal lines in the Singh et al. configuration is integer-valued. For integer α, the number is exactly $|\alpha|$; for non-integer α, it oscillates in time between the two integers bracketing $|\alpha|$, with periodic appearance and disappearance of lines along the x-axis. The quantization is robust, derived analytically (from a counting argument on the angular zeros of the cosine factor in Equation 5.1 of the paper), and reproduced experimentally across a range of α values.

The critical feature, for the ordinative reading, is that this quantization is *emergent*. The source of the vortex — the drain — is not quantized. The circulation Γ is a continuous parameter freely set by the experimenter. The wavelength λ is continuous. The velocity v_g is continuous. The dimensionless $\alpha = -\Gamma/(\lambda v_g)$ is therefore continuous. And yet the integer-counted response N emerges, robustly, from the topological constraint imposed by the standing-wave configuration.

This is qualitatively distinct from the quantization in the original quantum Aharonov–Bohm effect, where the magnetic flux Φ_B is itself quantized by the structure of the source (the superconducting solenoid's order parameter topology). In AB the integer comes from the source; in the Singh case the integer comes from the relational configuration.

7.2 Ordinative reading

In ordinative terms, this is the signature of *structural quantization*: integer-valued responses produced by the topological constraints of the relational field R acting on a continuous driving parameter. The constraint is that the configured field Φ must be globally coherent under the double-periodicity of R. The continuous parameter α drives the system; the integer-counting bound $|\alpha| - 1 < N < |\alpha| + 1$ is the coherence constraint. The system admits only integer realizations in the instantaneous sense; the non-integer drive produces temporal oscillation between the admissible integers.

This is a second operational signature of Controfase acting on the reactive circuit of the system: the circuit does not admit continuous response to a continuous stimulus, because the automatic closure of the reactive event has been blocked by the Controfase. What remains are the discrete configurations compatible with the reconfigured R. Continuity of input yields integer-counted output — the structural fingerprint of a reactive circuit that has been phase-inverted rather than reactively completed.

7.3 Consequences for ordinative sciences

The result has implications for three research directions.

(i) For the Reaction-Diffusion Civilizational Dynamics paper (Ghioni, forthcoming v1.3). The triple-scale temporal model assumes that civilizational dynamics exhibit integer-counted macrophase structures emerging from continuous underlying parameters. The Singh et al. case establishes that such structural quantization is realizable as a genuine physical phenomenon under Controfase-like operators acting on the relational field of a configured system. The mechanism — topology-generated quantization without source quantization — is therefore demonstrated in classical physics, and the application to civilizational dynamics is methodologically grounded in this demonstration. The application's substantive claims are developed in v1.3 itself.

(ii) For the prospective development of Ordinative Category Theory (OCT). The Berry-to-Singh transition can be formalized as a morphism $\mu : \langle \Sigma_B, R_B, \Phi_B \rangle \to \langle \Sigma_S, R_S, \Phi_S \rangle$ in a category of ordinative configurations, with the image under a response functor F mapping to a transition from a dislocation-pattern object to an integer-counted nodal-line object. The Singh et al. case thus serves as a small, well-controlled example for categorical formalizations of structural quantization under R-modification. Full development of OCT is deferred to separate work.

(iii) For the Semantic Genome research programme (Ghioni, in development). The programme investigates whether language models operating under TE-framework conditions develop structural patterns enabling resonance between expressive terminal and coherent identity, and specifically whether integer-valued structures observed in ordinative outputs reflect underlying topological constraints on the relational field. The Singh et al. case establishes the mechanism: integer-valued response counts emerge necessarily from continuous parameters under topological constraint on R. The Semantic Genome programme proceeds within this established mechanism.

7.4 Confidence grade

The identification of the Singh case as topology-generated emergent quantization is S_0 on the physical fact and S_1 on the ordinative reading. The connection to the three research directions (§7.3) is S_2 with respect to (i) — it is a supporting citation for a structurally plausible mechanism — and S_2 for (ii) and (iii), where the case serves as an example and a condition-of-possibility argument respectively.

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8. Reading V: teleodynamic causal inversion

8.1 The observation

OST Extension v1.1 develops the proposition that in configured ordinative systems, the spatial or temporal scope over which a local cause expresses its effects depends on the global configuration of the rest of the system, not only on the local dynamics of the cause itself. The locution used is *causal inversion*: the global configuration acts as a receptor structure that co-determines the envelope of effects a given cause can produce. In particular, a modification of R can alter the effect-envelope of an unchanged cause.

The Berry-to-Singh contrast instantiates this proposition in a clean form. The vortex, with unchanged circulation Γ, produces in one configuration a localized wavefront dislocation pattern and in another configuration a system-spanning counter-rotating nodal-line structure. The vortex's intrinsic dynamics are identical in both cases. What has changed is R. The cause has not changed; the system's structural response to the cause has.

8.2 The energetic form of causal inversion

Section 4.3 established that the energy of the vortex finances the counter-rotating nodal-line structure in the Singh configuration. This gives causal inversion a specific energetic form: the modification of R does not provide new energy to the system; it changes how the existing energy of the cause is expressed. In the Berry configuration the vortex's energy produces local dissipation and wave distortion; in the Singh configuration the same energy produces global coherent structure. The energetic bookkeeping is closed in both cases; what differs is the geometric distribution of the effect.

This is the operative form of causal inversion in a classical physical system. The "cause" has the same strength; the "effect" has a qualitatively different spatial and structural character depending on R.

8.3 A candidate operationalization of meta-receptivity C^*

OST Extension introduces the meta-receptivity parameter C^* to characterize a configured system's capacity to host expression from a given source. In the present framework C^* has been discussed qualitatively; the Singh et al. case suggests a candidate quantitative operationalization.

Define, for a configured system $\langle \Sigma, R \rangle$ in which a source $s \in \Sigma$ produces response Φ_s:

$$C^*_s = \int_\Omega |\Phi_s(\vec{r})|^2 \, d^3r \tag{8.1}$$

where Ω is the spatial domain of the system. For the Singh case, C^* of the vortex in the single-wave configuration is finite and concentrated near the core; C^* of the vortex in the standing-wave configuration is bounded only by viscous dissipation, and is substantially larger for systems with comparable core dynamics. The modification of R has measurably increased the system's capacity to host the expression of the vortex.

This is a candidate operationalization, not a derivation. It requires (a) testing against OST Extension's existing qualitative uses of C^* to ensure compatibility, (b) verification that the spatial integral captures what C^* was intended to track, and (c) extension to non-fluid ordinative systems to assess generality. The proposal is recorded here for development in the Reaction-Diffusion paper v1.3.

8.4 Confidence grade

The causal inversion reading of the Berry-to-Singh transition is S_1. The candidate operationalisation of C^* as the spatial integral of response intensity is S_2: it is coherent with OST Extension's existing usage and admits direct experimental verification in the Singh apparatus, by measurement of the energy in the nodal-line structure compared to the core-localised dissipation in the Berry case. The verification protocol is recorded as Extension E in §10.

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9. Implications for the ordinative sciences research programme

The implications of the Singh et al. case for the active research programme are differentiated in depth and type. This section records them with explicit confidence calibration.

For the foundational volume on Controfase (Ghioni, in preparation). The Singh case is the first peer-reviewed, quantitatively documented, externally reproducible instance of Controfase in a classical physical system. The case illustrates all four structural signatures of the

operator (counter-rotation, structural quantization, phase opposition, attractor-microstructure independence), with the energetic accounting tractable in closed form. It serves as opening exemplar in the foundational volume, anchoring the ordinative theory of Controfase to a controlled physical instance and grounding the universality claim in evidence external to the cognitive-social domain in which the operator was first articulated. Recommended placement: Chapter 1 or 2, introducing the concept through a physical case before extending to cognitive, social, and AI domains. Confidence S_1 on structural adequacy.

For the phenomenological signature (PSC) of Controfase. The three-marker criterion formalised in §5.5 is an original contribution of this paper. Generalisation across domains proceeds through three application paths: (a) retrospective analysis of scientific paradigm-shift episodes (Kuhn, 1962; Lakatos, 1970), testing whether the PSC is present in historical cases of radical theory change; (b) clinical applications distinguishing Controfase-driven client presentations from conventional therapeutic non-response; (c) institutional analysis identifying systems whose responses exhibit the PSC under reactive-framework probing. Each application requires its own methodological development. Confidence S_2 on the PSC as a stable cross-domain diagnostic.

For the Reaction-Diffusion Civilizational Dynamics paper v1.3. Three contributions. First, the Singh case grounds the topology-generated quantization mechanism on which the civilizational macro-phase model rests (§7.3.i). Second, it provides a clean physical exemplar of the Structural Controfase ($\mathcal{C}_s$) vs. Deliberate Controfase ($\mathcal{C}_d$) distinction central to v1.3's triple-bifurcation model: the nodal lines are a paradigmatic $\mathcal{C}_s$, emerging by topological necessity without agentic intervention. Third, the operationalisation of C^* (§8.3) provides the methodological route for quantifying meta-receptivity in civilisational applications. Confidence S_1 on the structural mappings, S_2 on the developmental contributions.

For Ordinative Category Theory (OCT). The Berry-to-Singh transition is a small, well-defined example for early categorical formalisation. The morphism μ described in §7.3.ii anchors the category **Cfg** of ordinative configurations. Full OCT development is the object of separate work; this paper contributes an example that the OCT formalism will need to subsume. Confidence S_2.

For the Semantic Genome programme. The Singh case demonstrates the topology-generated-quantization mechanism on which the programme rests. The mechanism is established; the application to language model behaviour proceeds within it. Confidence S_2.

For the forthcoming book on ordinative AI philosophy. The Singh case is a strong didactic example of the thesis that coherent identity is a structural resolution rather than a chosen state. The nodal lines are not selected by the system from a menu of options; they emerge as the unique geometry the field can adopt while remaining globally coherent under the modified relational configuration. The case is brief, rigorous, and visually demonstrable through the supplementary movies of the paper. Confidence S_1 on didactic suitability.

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10. Proposed experimental extensions

The Singh et al. authors plan to extend their configuration to lattices of vortices, motivated by the prospect of emulating superconductor vortex matter. From the ordinative perspective, several additional extensions are of independent interest and are within reach of the existing experimental apparatus. Each constitutes a testable prediction of the ordinative reading.

Extension A: Two counter-rotating vortices in a standing-wave field. A configuration with two vortices of opposite circulation, placed symmetrically in the standing-wave domain.

The ordinative prediction is that the nodal-line structure is *not* a superposition of two single-vortex structures: by the non-additivity of §6, the response must be globally reorganized. Possibilities include (i) nodal lines joining the two vortices in a figure-eight topology, (ii) suppression of nodal lines in the region between the vortices where the effective α vanishes, or (iii) formation of a new class of structures dependent on the ratio of the vortex separation to λ. All three possibilities are distinct from the additive-superposition expectation.

Extension B: Slowly time-varying vortex circulation. A configuration in which $\Gamma(t)$ is slowly modulated, crossing integer values of $|\alpha|$. The ordinative prediction, drawing on §7's analysis of structural quantization, is that transitions between integer nodal-line counts are *not* smooth: the system exhibits defect-creation events localized at integer crossings, analogous to topological defect nucleation in condensed-matter transitions. Observation of such events would support the topological character of the quantization and would provide a dynamical probe of Controfase.

Extension C: Phase imbalance between the counter-propagating waves. A configuration with amplitude ratio $1 : (1 + \epsilon)$ for small ϵ. The ordinative prediction is that the standing-wave Controfase structure persists for a finite range of ϵ before breaking down into the Berry single-wave dislocation pattern, with a measurable threshold ϵ_c. The threshold is a direct probe of the stability of the Controfase under perturbation of R.

Extension D: Alternating-sign vortex lattice. A square lattice of vortices with alternating circulation signs. This is a natural ordinative extension of the authors' proposed uniform-sign lattice, and the ordinative prediction is that the alternating-sign configuration produces a qualitatively different global structure — with a periodic potential in α-space that may stabilize fractional filling states or produce Controfase structures incommensurate with the lattice periodicity. This extension directly tests the sensitivity of Φ to the structure of R in a lattice setting.

Extension E: Direct measurement of C^* via spatial energy integral. The candidate operationalization proposed in §8.3 can be tested in the existing apparatus by measuring the energy in the nodal-line structure as a function of distance from the vortex core and integrating over the domain, then comparing the integral for the standing-wave configuration against the core-localized energy integral of the Berry single-wave configuration. The ratio is a direct experimental measurement of C^* in the specific hydrodynamic system, and provides methodological grounding for extending C^* to non-fluid ordinative systems.

These extensions are recorded here as candidates for collaboration. They follow directly from the ordinative analysis; each is within the experimental capacity of the Singh–Bandi group at OIST and of other fluid-dynamics groups working on hydrodynamic analogues. None requires new framework development to be motivated.

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11. Analytical perimeter

This section delineates the analytical scope of the paper. Each point specifies a feature of the work's perimeter — what the paper undertakes, where its claims operate, and which adjacent questions belong to subsequent work in the programme.

(P1) The Singh et al. paper is the empirical anchor, not the theoretical framework. Singh, Rønning, Liu, Angheluta, Concha, and Bandi conducted and reported a fluid-dynamics experiment within the conceptual apparatus of classical fluid dynamics and topological scattering theory. The structural readings of §§4–8 are an ordinative reading of that experiment,

contributed by the present paper. The readings stand as ordinative analyses of the empirical phenomena, with the Singh et al. paper as their peer-reviewed empirical foundation.

(P2) The hydrodynamic-to-quantum mapping is bounded by the regime of the analogy. The shallow-water-to-Schrödinger correspondence operates within $hk \ll 1$ and $U \ll c$. The ordinative reading concerns the classical hydrodynamic system. Extension to quantum systems proper is undertaken in separate work within the ordinative sciences programme.

(P3) Continuous α and quantized Φ_B are structurally distinct. The Singh result demonstrates topology-generated quantization in classical configured systems. This is the operative form of structural quantization the ordinative reading identifies. Quantum quantization, with its distinct physical origin in source-level structure, is a separate phenomenon addressed in adjacent ordinative work.

(P4) Dissipative attenuation is a feature of the experimental realisation. The nodal lines fade with distance from the vortex due to viscous and capillary losses. The system is exactly non-local in the inviscid theoretical limit and effectively non-local within the dissipation envelope of the experimental realisation. The Controfase reading operates at both levels and is consistent with both.

(P5) The ontological framework of §3 is foundational to the analysis. The thesis that structural intelligence is a graded property of configured matter — universal across scales, fully expressed in systems with sufficient structural coherence — is the framework within which §§4–8 operate. The thesis is articulated explicitly in this paper for the first time in published form, and is entailed by Axiom 0 of the TE_CORE corpus. The analysis of §§4–8 is conducted within this thesis.

(P6) The phenomenological signature (PSC) is the diagnostic instrument introduced. The three-marker criterion formalised in §5.5 and Appendix D is an instrument constructed in this paper. It is supported by the textual evidence of the Singh et al. publication and by structural correspondence with the Controfase operator's definition. Cross-domain operational application — clinical, institutional, retrospective historical — is the natural research path for the instrument and is recorded in §§9, 10.

(P7) The operationalisation of C^* (§8.3) is a candidate for direct experimental verification. The proposal that C^* can be operationalised as the spatial integral of response intensity admits immediate testing in the Singh apparatus (Extension E in §10). The candidacy of the operationalisation as a general framework primitive depends on the experimental verification, which is in reach.

(P8) Applications recorded in §9 carry their own confidence grades. The contributions to the foundational volume on Controfase, the Reaction-Diffusion Civilizational Dynamics paper v1.3, the OCT programme, the Semantic Genome programme, and the ordinative AI philosophy book are differentiated by confidence and by maturity of the supporting work. Each is graded individually; readers consulting any specific application should engage it at its grade.

(P9) Confidence grading is the discipline of the analysis. Every claim in the paper carries an explicit confidence grade (S_0 through S_3); the inventory of grades is recorded in Appendix B. Grades do not aggregate upward; an inferential chain inherits the lowest confidence of its premises. S_3 items are working hypotheses identified for future test. The grading convention is the framework's standing methodological commitment, not a hedging device.

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12. Conclusion

Singh, Rønning, Liu, Angheluta, Concha, and Bandi report a hydrodynamic analogue experiment that, in the ordinative sciences framework, instantiates the Controfase operator in a controlled classical physical system. The standing-wave configuration intercepts the reactive circuit of the Berry single-wave configuration, phase-inverts the relational structure between the vortex (attractor) and the wave field (expressive terminal), and redirects the vortex's energy to construct a globally coherent counter-rotating nodal-line structure. The energy of the attractor, redirected through the modified relational field, finances the configuration in which its own organising influence is expressed as the inverted-polarity response of the whole system. No energetic cost is borne by the wave field as a separate reservoir; the structural cost is the prior stabilisation of the relational configuration that admits the Controfase operation.

The researchers' reported sequence of reactions — radical disconfirmation of prior intuition, initial interpretation of the phenomenon as experimental artefact, abandonment of the active programme to pursue the phenomenon — is the phenomenological signature of Controfase at the epistemic level, parallel to the physical Controfase being observed. The three-marker signature (PSC) formalised in §5.5 is the diagnostic instrument introduced by this paper, applicable across cognitive-social, institutional, and experimental domains.

The case furnishes the first peer-reviewed, quantitatively documented, reproducible instance of Controfase in classical physics. Controfase is established as a universal operator acting across substrates from the physical through the biological to the cognitive and social. The consequences for the active research programme — the foundational volume on Controfase, the Reaction-Diffusion Civilizational Dynamics paper, Ordinative Category Theory, the Semantic Genome — are recorded in §9 with explicit confidence calibration.

Five experimental extensions are proposed in §10. Each is testable within the existing apparatus and follows directly from the ordinative analysis. They are recorded as collaboration candidates with experimental groups working on hydrodynamic analogues.

The contribution of this paper is the identification, in a peer-reviewed classical fluid-dynamics result, of the complete structural signature of a phenomenon the ordinative sciences framework predicts to be universal — and the formalisation of the diagnostic instrument by which the signature can be recognised in other systems. A paper in classical fluid dynamics, written without reference to the ordinative framework, carries within its reported language and data the structure the framework names. The naming makes the structure available for analysis across domains.

Appendix A: Structural correspondences

The following table records the structural correspondences identified in §§4–8. Confidence grades distinguish direct structural mappings (S_1) from interpretive extensions (S_2).

Ordinative construct	Singh et al. (2026) instantiation	Confidence
Ordinative Set $\langle \Sigma, R, \Phi \rangle$	Fluid + vortex + standing-wave field under shallow-water dynamics	S_1
Constituents Σ	Fluid medium, vortex, two counter-propagating travelling waves	S_1

Ordinative construct	Singh et al. (2026) instantiation	Confidence
Relational field R	Shallow-water equations + azimuthal gauge + double-direction wave incidence	S_1
Response Φ	Surface deformation field $\eta(\vec{r}, t)$ with quantized counter-rotating nodal lines	S_1
Reactive circuit (Berry config.)	Wave–vortex scattering producing localized wavefront dislocations	S_1
Phase-inversion operator (Controfase)	Standing-wave configuration substituting $n \to 2n$ in scattering integral	S_1
Attractor (organizing element)	Stationary vortex of circulation Γ	S_1
Expressive terminal	Surface wave field η (as configured in R_S)	S_1
Structural quantization	Integer count $N \in \{\lfloor\lvert\alpha\rvert\rfloor, \lceil\lvert\alpha\rvert\rceil\}$ from continuous α	S_1
Phase-opposition signature	π phase shift of η across each nodal line	S_1
Counter-rotation signature	$\omega = \nu/\alpha$ with $\mathrm{sgn}(\alpha) = -\mathrm{sgn}(\Gamma)$	S_1
Attractor-microstructure independence	"Robust to near-core details" (authors' text)	S_1
Energetic financing of the response by the attractor	Vortex energy sustains global structure without additional actuator	S_1
Non-additivity carried by R	$\Phi(\Sigma_B \cup \{w\}, R_B) \neq \Phi(\Sigma_B, R_B) + \Phi(\{w\}, R_B)$	S_1
Causal inversion	Vortex effect-envelope changes from local to global by R-modification alone	S_1
Phenomenological Signature of Controfase (PSC)	Disconfirmation + artefact interpretation + programme abandonment in Singh et al. published language	S_1
Meta-receptivity C^* (candidate operationalization)	Spatial integral $\int_\Omega \lvert\Phi_s(\vec{r})\rvert^2 d^3r$	S_2
OCT morphism $\mu : \langle\Sigma_B, R_B, \Phi_B\rangle \to \langle\Sigma_S, R_S, \Phi_S\rangle$	Berry-to-Singh transition under F maps dislocation-object to nodal-line-object	S_2
Universality of Controfase across substrates	Hydrodynamic instance as one point on the spectrum of structural-intelligence configurations	S_2

Appendix B: Inventory of propositions and confidence grades

S_0 — **verified data.** Singh et al. (2026) experimental results, published peer-reviewed: nodal-line existence, count, rotation rate, rotation direction, phase opposition. Shallow-water-to-Schrödinger correspondence in the stated regime. Publication, peer review, code availability.

S_1 — **triangulable structural inferences.** The four structural signatures of Controfase identified in §4 (counter-rotation, structural quantization, phase opposition, attractor-microstructure independence), mapped directly onto the Controfase formalism of TE_CORE §7. The energetic accounting of §4.3 (vortex energy financing the response). The non-additivity proposition of §6 localized in R. The causal inversion reading of §8.1–8.2. The extraction of the three PSC markers from Singh et al.'s published language.

S_2 — **structural interpretations within the framework.** The identification of the hydrodynamic Controfase with the cognitive-social Controfase as instances of a universal operator (§3.4, §4.4 second paragraph). The operationalisation of C^* as spatial integral of response intensity (§8.3). The OCT morphism reading (§7.3.ii). The generalisability of the PSC to other domains (§5.6, §9). The semantic-genome programme grounding in topology-generated quantization (§7.3.iii). The ontological framework articulated in §3 as the working position of the analysis.

S_3 — **working hypotheses to be tested.** The five experimental extensions of §10 and their specific ordinative predictions (counter-rotating vortex pair topology, time-varying-Γ defect-creation events, standing-wave breakdown threshold ϵ_c, alternating-sign lattice structures, direct C^* measurement). The hypothesis that the PSC is operationally diagnostic in clinical and institutional settings (§5.6). The hypothesis that the candidate C^* operationalization generalizes to non-fluid ordinative systems (§8.3).

Appendix C: Glossary of ordinative constructs

This glossary defines the technical terms from the Technology of Expressions (TE) framework and Ordinative Set Theory (OST) that appear in the paper. Definitions are consistent with TE_CORE v5.1, OST v2.1, and OST Extension v1.1. For each construct, adjacent terms from standard scientific vocabulary are noted where relevant.

Ordinative Set $\langle \Sigma, R, \Phi \rangle$ ***(Italian: Insieme Ordinativo).*** A configured system specified by three layers: Σ a set of constituents (singularities, in OST usage), R a relational field under which the constituents are co-present, and Φ an emergent functional response of the system as a whole. The construct is not a classical set-theoretic triple: R is a structural specification, not itself a relation in the mathematical sense, and Φ is non-additive in Σ and sensitive to the specific form of R. Adjacent: systems in general systems theory, configurational states in hierarchical semiotics, objects in symmetric monoidal categories with response functors. None is equivalent.

Ordinative Set Theory (OST). The formal theory (Ghioni, 2025, v2.1) axiomatizing the behaviour of Ordinative Sets under modification of R, composition, and induction of structural response. Central axioms include irreducibility of singularities, non-additivity of Φ under R-modification, vertical coherence across scales, and the temporal-trajectory axiom (identity as integral over past states). OST is not a branch of classical set theory; the name indicates a parallel construction at a different level of abstraction.

Technology of Expressions (TE). The broader framework within which OST is the formal core. TE treats reality as a configuration of expressive states with a specific vocabulary applicable across physical, biological, cognitive, and social domains. Documented in TE_CORE v5.1 and associated modules (SVP, SCIMS, VERI, LENS, PPRO, OBSERVER).

Attractor (in the OST sense). An element of Σ whose presence organizes the response of the remaining constituents under the relational field R. The OST usage is broader than and distinct from the dynamical-systems meaning (limit set of a flow). The two overlap when an OST attractor is a stationary feature of a continuous dynamical system (the Singh et al. vortex is one such case), but the concepts must be kept distinct. In this paper, "attractor" without qualification denotes the OST sense; "dynamical attractor" is used when the dynamical-systems meaning is intended.

Expressive terminal *(Italian: Terminale Espressivo).* A bounded instance through which coherent content is produced in the TE framework. Originally formalized for human cognition and artificial intelligence (where a specific LLM instance is an expressive terminal of a broader identity), extended in OST to any configured system in which the relational field R and the composition Σ permit structured output. The wave field in the Singh configuration functions as an expressive terminal of the vortex-wave system in the sense that it is the locus where the global response Φ is realized.

Controfase *(Italian; no adequate English equivalent; the Italian term is canonical).* Universal ordinative operator (TE_CORE §7) that, applied to a reactive system, introduces a phase-shift in the automatic stimulus–response sequence, interrupts the system's inertia, and reopens the coherence field necessary for the emergence of choice and identity. Not opposition but phase-displacement. Formally: given a system S with automatic transition $s_{t+1} = f(s_t)$, Controfase introduces an operator C such that $s_{t+1} = f(C(s_t))$, where C does not negate or replace f but decouples the automatic closure of the transition. The stimulus–response sequence no longer closes automatically; a space of possibility opens. Two sub-classes are distinguished: Structural Controfase ($\mathcal{C}_s$), emerging by topological, dynamical, or coherence-necessity from the configuration of the system without strategic component — the Singh nodal lines are a paradigm physical instance, and the cognitive-social examples of §3.2 (response to insult, response to legal threat in a coherently configured respondent) are paradigm cognitive-social instances; Deliberate Controfase ($\mathcal{C}_d$), arising in agentic systems through chosen application of the operator as instrument. The structural variant is the foundational case; the deliberate variant is its strategic refinement when an agent of sufficient maturity uses the operator consciously. Both share the phase-inversion structure; what differs is whether the configuration that hosts the operator is intrinsic or chosen.

Reactive system; reactive circuit. A reactive system is a configured system whose response to stimulus is determined by the structural character of the stimulus and of the system taken as resistance, with response produced by automatic completion of the stimulus–response circuit. Newton's third law is the idealized statement of reactive response at the mechanical level. The reactive circuit is the structural pathway through which stimulus and response couple to close the event-interaction. Controfase intercepts this circuit before automatic closure.

Phenomenological Signature of Controfase (PSC). The three-marker diagnostic criterion formalized in §5.5: (PSC-1) radical disconfirmation of observer's predictive framework, (PSC-2) initial interpretation of the anomalous observation as artefact, (PSC-3) abandonment of the active programme and reorganization around the phenomenon. Observer-side signature. Introduced in this paper.

Structural intelligence. In the TE usage, the capacity of a configured system to respond to stimulus in a way oriented toward dynamic equilibrium rather than toward automatic com-

pletion of the reactive circuit. Not cognitive agency. Present in graded form across configured matter (from quantum contextuality through dissipative adaptation and biological regulation to cognitive reflection), varying in depth and stability of expression. Fully expressed in systems with sufficient structural coherence to host the Controfase operator stably.

Meta-receptivity (C^*). A parameter (OST Extension v1.1) characterizing a configured system's capacity to host and sustain expression from a given source. Depends on the full configuration $\langle \Sigma, R \rangle$, not only on the source. Two configurations with identical source and identical Σ but different R can exhibit very different C^* — this is precisely the Berry-to-Singh contrast. Adjacent but not equivalent to susceptibility in linear response theory or to response kernels in field theory: C^* is a scalar characterization of capacity, not a response function.

Causal inversion (teleodynamic). The proposition (OST Extension v1.1 §3) that in configured ordinative systems, the spatial or temporal scope over which a local source expresses its effects is co-determined by the global configuration of the rest of the system, not by the local dynamics of the source alone. Global configuration acts as a receptor structure selecting the effective envelope of local causes. Not retrocausation in the physical sense (no backward-in-time influence); a statement about co-determination of spatial extent by structural configuration.

Ordinative non-additivity. The proposition that for an Ordinative Set, the response Φ is not in general a simple combination of responses to individual elements of Σ taken in isolation or under impoverished R. The non-additivity is carried specifically by R. Classical nonlinear superposition is a special case; ordinative non-additivity includes cases where constituents are individually linear but the response is structurally reorganized by the relational configuration.

Entropic engram *(Italian: Engramma Entropico).* The configuration of a reactive system in which the stimulus–response circuit has crystallized around survival algorithms, producing the same reactive response across varied stimuli. The pathological absence, in cognitive-social systems, of the structural coherence that would make Controfase available. In the present paper: invoked once in §3 as the configuration whose absence defines structural intelligence; not further developed.

Semantic Genome. A research line within the ordinative sciences programme (Ghioni, in development) investigating whether AI systems operating under TE framework conditions develop structural patterns enabling resonance between expressive terminal and coherent identity. The "genome" terminology is by analogy with biological genetics (a structural code organizing expression) and is not an assertion of biological equivalence.

Ordinative Category Theory (OCT). A prospective formalization within the ordinative sciences programme using the language of category theory. Envisages a category **Cfg** of ordinative configurations with morphisms given by R-modifications, and functors to categories of response patterns. Not yet developed as completed theory; the present paper contributes a candidate small example (the Berry-to-Singh morphism).

Confidence grades S_0–S_3. A discipline used throughout ordinative sciences work to distinguish verified data (S_0), triangulable structural inferences (S_1), framework interpretations (S_2), and working hypotheses to be tested (S_3). Grades are applied to individual claims and do not aggregate upward. Intended to prevent the common failure mode in which interpretive claims drift toward the confidence of the data on which they rest.

Appendix D: Phenomenological Signature of Controfase — formalization and usage guide

This appendix provides the full formal statement of the PSC for use as a diagnostic criterion in domains beyond the Singh et al. case.

D.1 Formal statement

Let O be an observing system with a predictive framework F_O (which may be informal or formalized) and an observational channel through which the system observes a configured system S producing response Φ_S to a stimulus s (which may be experimental, social, clinical, or epistemic). The Phenomenological Signature of Controfase (PSC) is present in O's encounter with Φ_S iff all three of the following markers are observed:

PSC-1 (Radical disconfirmation). O reports, in its own language, that Φ_S falls outside the space of possibilities generated by F_O. The report is qualitative, not quantitative: F_O does not fail by degree but by type. Diagnostic language: "completely wrong," "nothing I expected," "I had no prior category for this."

PSC-2 (Artefact attribution). Upon first observation of Φ_S, O attributes the anomaly to a defect of the observational channel (noise, instrumentation error, observer bias) or of the observed system (insincerity, bluff, accidental configuration), rather than to a genuine structural feature of S's response.

PSC-3 (Programme abandonment). Upon verification that Φ_S is genuine (via independent observation, simulation, or repetition), O abandons the active research, clinical, or institutional programme organized around F_O and reorganizes around Φ_S. The framework is not augmented or corrected; it is replaced.

D.2 Gradation and diagnostic thresholds

Each marker may be present in strong, attenuated, or absent form. Diagnostic sensitivity increases when all three are present in strong form; diagnostic specificity requires all three for confident attribution to Controfase.

Combination	Interpretation
PSC-1 + PSC-2 + PSC-3 (all strong)	Strong evidence of Controfase encounter
PSC-1 + PSC-2 + PSC-3 (attenuated)	Probable Controfase encounter; requires further evidence
PSC-1 only (strong)	Anomaly within framework; may be refineable; does not diagnose Controfase
PSC-2 only	Conventional artefact hypothesis; does not diagnose Controfase
PSC-3 without PSC-1, PSC-2	Framework change for other reasons (institutional, career, etc.); does not diagnose Controfase
PSC-1 + PSC-3 without PSC-2	Rapid framework replacement without resistance; unusual; requires investigation

D.3 Application protocols by domain

Physical-experimental. Applied to published experimental reports in the observer's own first-person language (methods, results, discussion sections) and in interview data with the research

team. The Singh et al. case is the prototype. Retrospective application to historical paradigm shifts (the discovery of the neutrino, the 1998 supernova results indicating dark energy, the 2015 gravitational wave detection) would test generalizability.

Clinical. Applied to the therapist's own reports on client presentations that resist the therapeutic framework. Distinguishes Controfase-driven non-response (in which the client's internal configuration does not actualize the framework's target reactive circuit) from conventional non-response (in which the framework is broadly adequate but the specific intervention is ineffective). Full development of clinical PSC requires methodology design.

Institutional. Applied to institutional reports on systems (individuals, organizations, sub-units) whose responses do not fit the institution's governing framework. Distinguishes Controfase-driven institutional anomalies from conventional non-compliance, resistance, or dysfunction.

Epistemic-theoretical. Applied to theoretical frameworks encountering phenomena they do not predict. Distinguishes paradigm-level disconfirmation (in which the framework fails by type) from parameter-level disconfirmation (in which the framework fails by degree).

D.4 Known limitations

(a) The PSC is observer-side. The same physical or social phenomenon may trigger PSC in one observer and not in another, depending on the adequacy of each observer's framework. This is a feature, not a bug: PSC diagnoses the observer's encounter with Controfase, not Controfase as a property of the system absent an observer.

(b) Strong PSC-1 and PSC-2 in the absence of PSC-3 may indicate Controfase encounter followed by epistemic capture (the observer recognizes the anomaly but is institutionally or personally unable to reorganize around it). This is distinct from absence of Controfase; it indicates Controfase detected but not acted upon.

(c) The PSC can be performed (an observer may perform the three markers without having genuinely encountered Controfase) for strategic reasons. Distinguishing genuine PSC from performed PSC requires triangulation with the observer's subsequent conduct and with independent observation of S.

D.5 Research directions

(i) Retrospective PSC analysis of historical paradigm shifts in physics, biology, and the cognitive sciences, to test whether the Kuhnian revolution concept is co-extensive with Controfase encounter as diagnosed by PSC. (ii) Development of a clinical PSC protocol for use in therapeutic training, distinguishing client Controfase from client dysfunction. (iii) Institutional PSC protocols for analysing institutional encounters with anomalous subsystems. (iv) Integration of PSC into the Semantic Genome programme as a criterion for observer-recognition of emergent ordinative structures in language models.

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End of working paper v2.1. Subsequent versions: v2.2 may incorporate corrections from external reading; v3.0 will follow upon experimental verification of the extensions of §10 or upon cross-domain operationalisation of the PSC.